

Dana Martinez Gonzalez

Mechanical Engineering | Aerospace Engineering

Engineering portfolio

School Fresno State	Program BS/MS Mech. Eng.	Minor Aerospace	GPA 3.52
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About me

I'm a mechanical engineering student at Fresno State in the Finish in Five program, working toward my BS with an aerospace minor (Spring 2027) and my MS (Spring 2028), with a 3.52 GPA. I started at West Hills Community College, where I earned my AA, before transferring to Fresno State's Lyles College of Engineering. My coursework and projects focus on structural analysis, design, and testing, and I've applied that work through three NASA internships.

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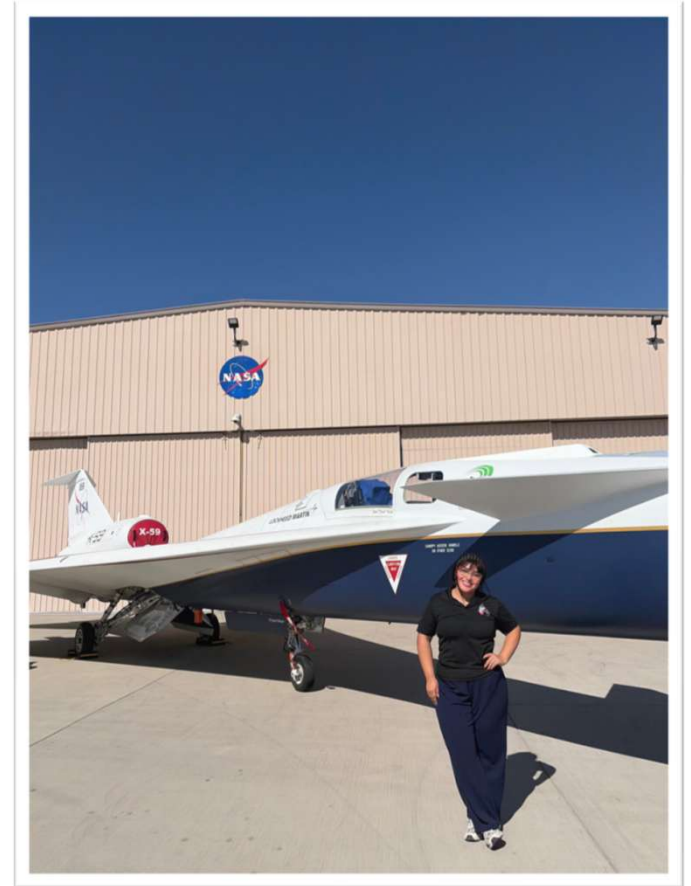
NASA internships

3.52

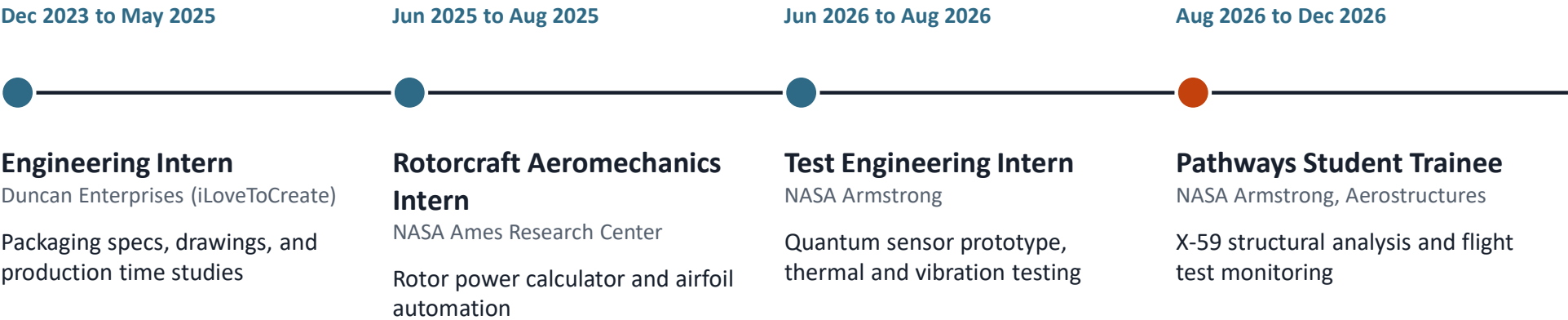
GPA

2028

MS expected



Experience



Also: ME 112 Dynamics grader at Fresno State, Fall 2026

Projects

Analysis, design, and test work from NASA, industry, and school

X-59 QueSST structural analysis

NASA Pathways, Armstrong Flight Research Center, Fall 2026

- Supporting structural analysis and stress evaluations in the Aerostructures Branch
- Monitoring real-time flight test data in the control room to catch anomalies
- Working with Lockheed Martin engineers on structural and flight test data
- Writing Python scripts to process flight test and structural data faster
- Modeling structural components in SolidWorks and PTC Creo for design reviews

FEA

Python

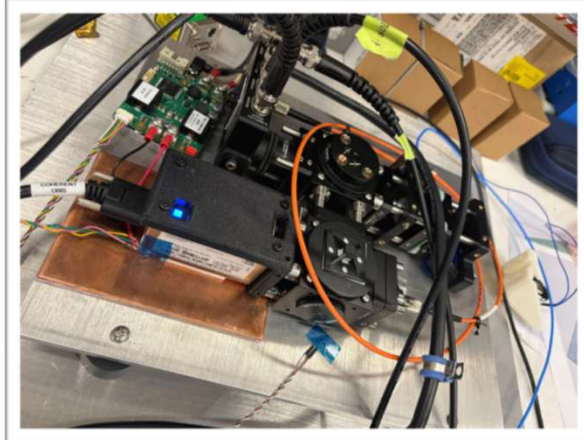
SolidWorks

PTC Creo

Flight test



Quantum sensor prototype



Test Engineering Intern, NASA Armstrong, Summer 2026

- Helped develop a sensor prototype combining RF, optics, and PCB hardware with software
- Ran thermal tests in a Thermotron chamber across the operating temperature range
- Ran vibration tests under flight-representative dynamic loading
- Designed TEC controller logic to keep the laser temperature stable
- Supported mechanical design and integration with a multidisciplinary team

Thermal testing

Vibration testing

TEC control

RF

PCB

CAD

Rotor power calculator and airfoil slicing

Rotorcraft Aeromechanics Intern, NASA Ames, Summer 2025

- Built a Python rotor power calculator with a custom GUI using interpolated CAMRAD II power coefficient tables
- Automated airfoil slicing from 3D rotor blade geometry in Rhino with rhinoscriptsyntax
- Supported next-generation vertical flight research using CFD, aeroelastic analysis, and wind tunnel testing

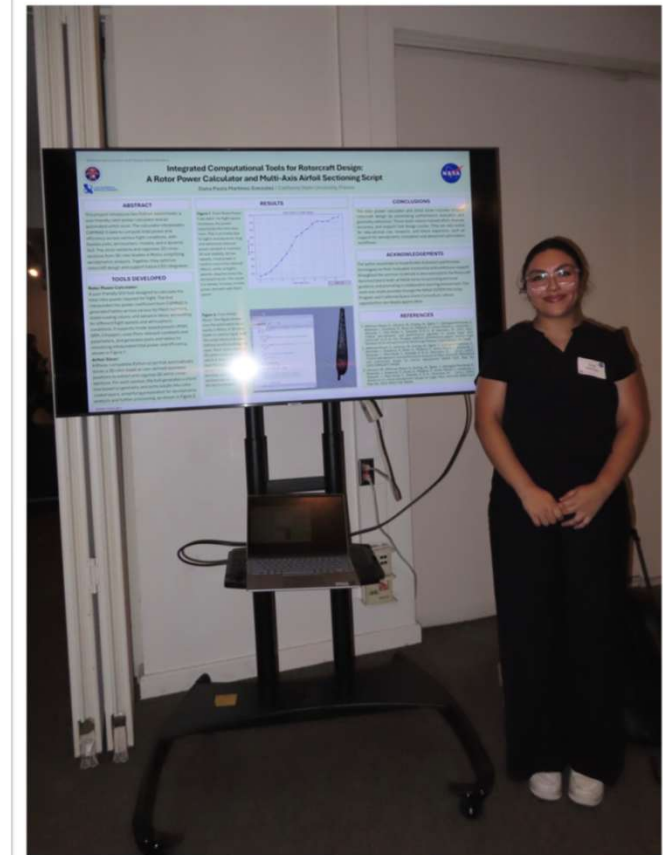
Python

GUI

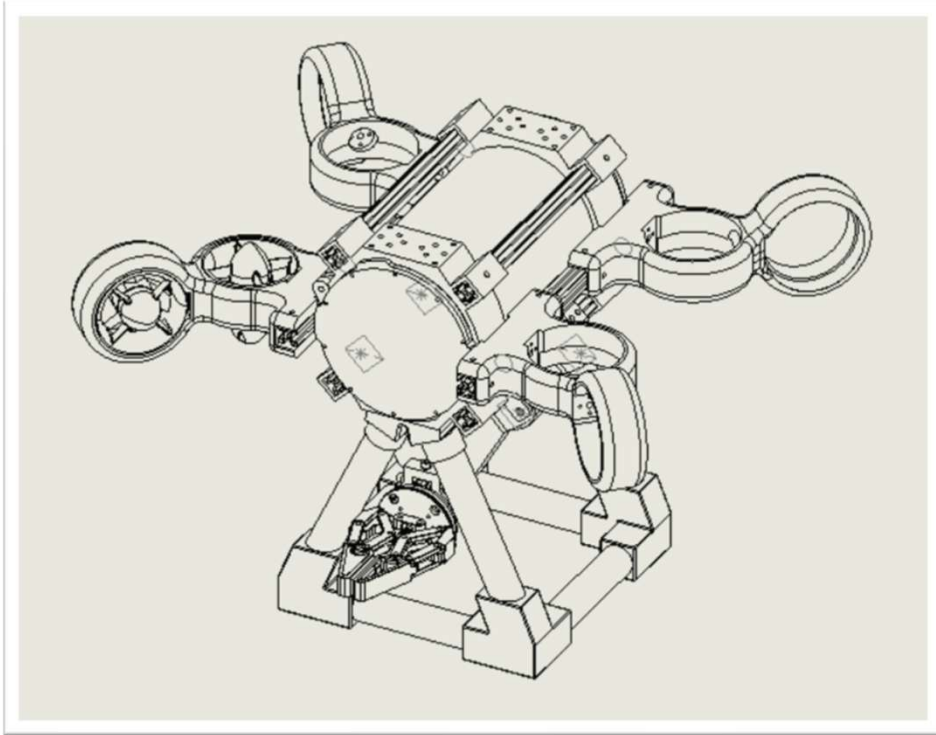
CAMRAD II

Rhino

CFD



MATE ROV chassis and propulsion



Fresno State MATE ROV team, November 2025 to present

- Designed and evaluated a six-thruster layout for full 6 DOF control and station keeping in flowing water
- Worked on chassis layout, thruster placement, and shrouds for stability, maneuverability, and safety
- Developed tether routing and strain relief concepts to protect onboard electronics

CAD

Propulsion layout

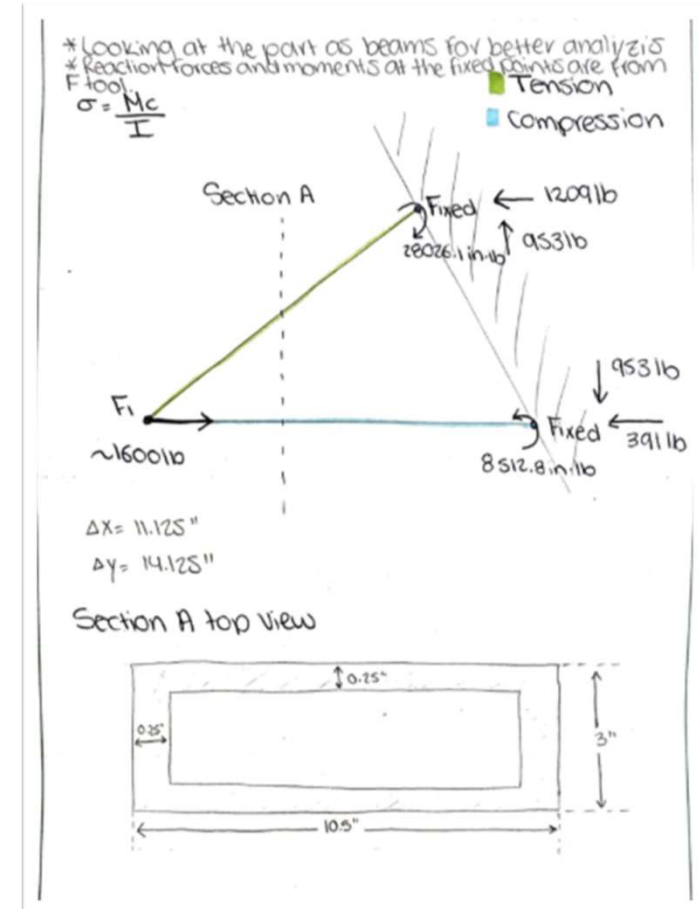
6 DOF control

Team design

Aircraft engine mount bracket analysis

NASA Pathways Internship, Fall 2026

- Analyzed an existing engine mount made of two mirrored triangular gusset links in 6061 aluminum
- Checked against design standards that it carries about 600 lbf without failing or deflecting over time
- Focused on the stress concentration at the rectangular clearance cutout in each link's web



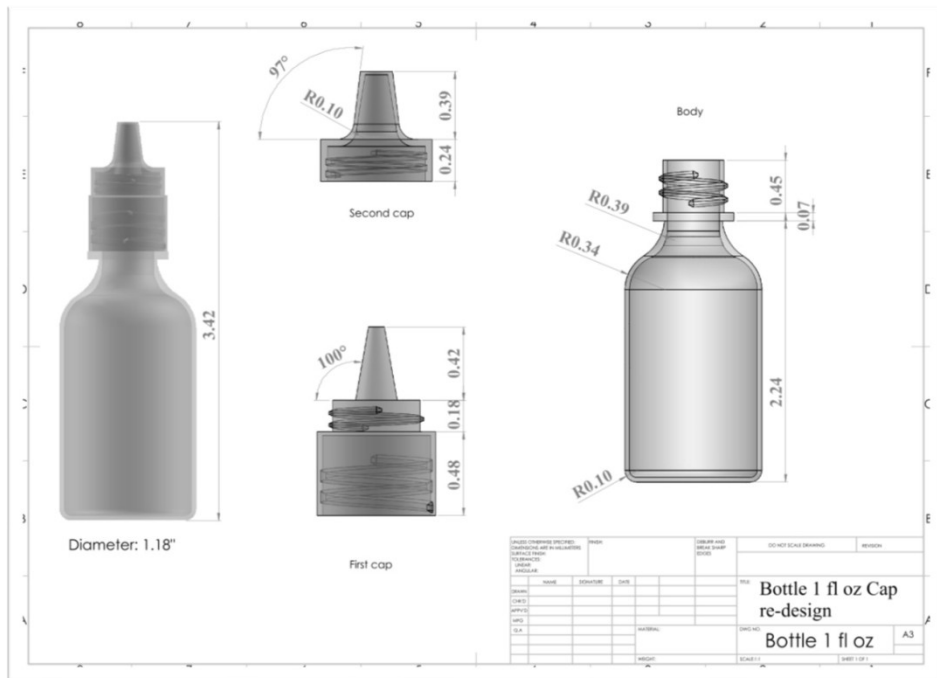
Stress analysis

FEA

Hand calcs

6061 aluminum

Packaging specs and production improvements



Engineering Intern, Duncan Enterprises, Dec 2023 to May 2025

- Created packaging specifications and engineering drawings for new product lines in SolidWorks and ArtiosCAD
- Ran time studies and data analysis to support production improvements and efficiency reporting
- Wrote standardized packaging instructions with Operations and Quality leadership

SolidWorks

ArtiosCAD

Time studies

Drawings

Skills

Engineering and analysis

Systems engineering

CFD

FEA

Numerical methods

MTS testing

Thermotron chambers

Controls

Software and tools

SolidWorks

PTC Creo

Siemens NX

ANSYS

MATLAB

Simulink

Python

AutoCAD

Revit

SketchUp

ArtiosCAD

Hardware and systems

RF systems

PCB design

Arduino

Soldering

TEC controllers

Data acquisition

Instrumentation

Programming

Python

MATLAB

C++

Manufacturing

CNC machining

Manual machining

Leadership and community

President, Society of Women Engineers, Fresno State

June 2026 to May 2027

- Lead seven-chapter officers across programming, professional development, and K-14 outreach
- Direct the WE26 national conference sponsorship campaign: corporate outreach, a tiered sponsorship packet, and officer task assignments to fund student travel to Boston

SHPE

Former Secretary

AIAA

Member

Mentor

First-year engineers



Education

- **MS Mechanical Engineering**
Fresno State, Finish in Five program
Expected Spring 2028
- **BS Mechanical Engineering, Aerospace minor**
Fresno State, Lyles College of Engineering, GPA 3.52
Expected Spring 2027
- **AA Liberal Arts and Humanities**
West Hills Community College
2022



Thank you

Open to internships and full-time roles in structures, propulsion, testing,
and hardware.

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